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# Analysis of the Movie Industry & Profitability Prediction

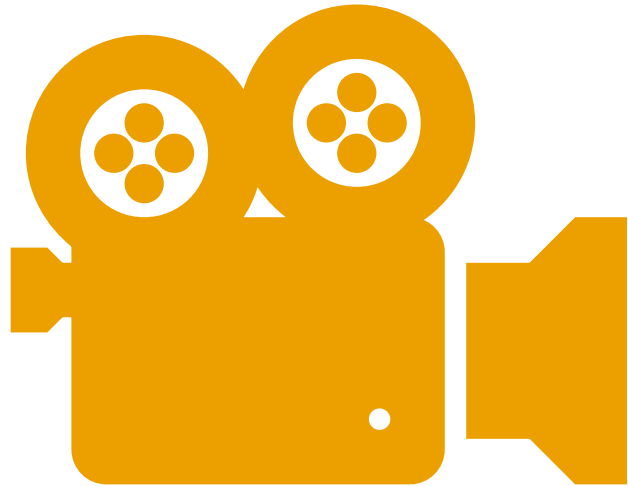
*MKT 568 – Marketing Analytics*

*BY Group Members:*

*Vaishnavi Chandalada*

*Shahid Baig Shaik*

# Project Objective



**What Was the Goal?**



Analyze 5,400+ movies from IMDb (1980–2025)



Identify key drivers of movie profitability



Apply:

Descriptive analytics

Regression (hypothesis testing)

Predictive modeling



Predict profitability for future movie releases



**Key Question:**

*What actually makes a movie profitable?*

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# Dataset Overview

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- **Data Description**
  - Source: IMDb (via Kaggle)
  - Final dataset: 5,410 movies
  - Original variables: 14
  - Engineered features: 10
  - Key variables:
    - Budget (inflation-adjusted)
    - Gross revenue
    - Profit (Gross – Budget)
    - Votes, Score, Genre, Rating
    - Release timing (month, season, day bin)

# Methodology

## Analytical Approach

## Data Cleaning

- Removed 14 records with invalid release dates
- Retained extreme values (real industry outliers)

## Feature Engineering

- Inflation-adjusted budgets (3.09%)
- Profit computation
- Dummy encoding of categorical variables

## Analysis Techniques

- Descriptive statistics & correlations
- Multiple Linear Regression
- Decision Tree & Neural Network models

# Key Descriptive Insights

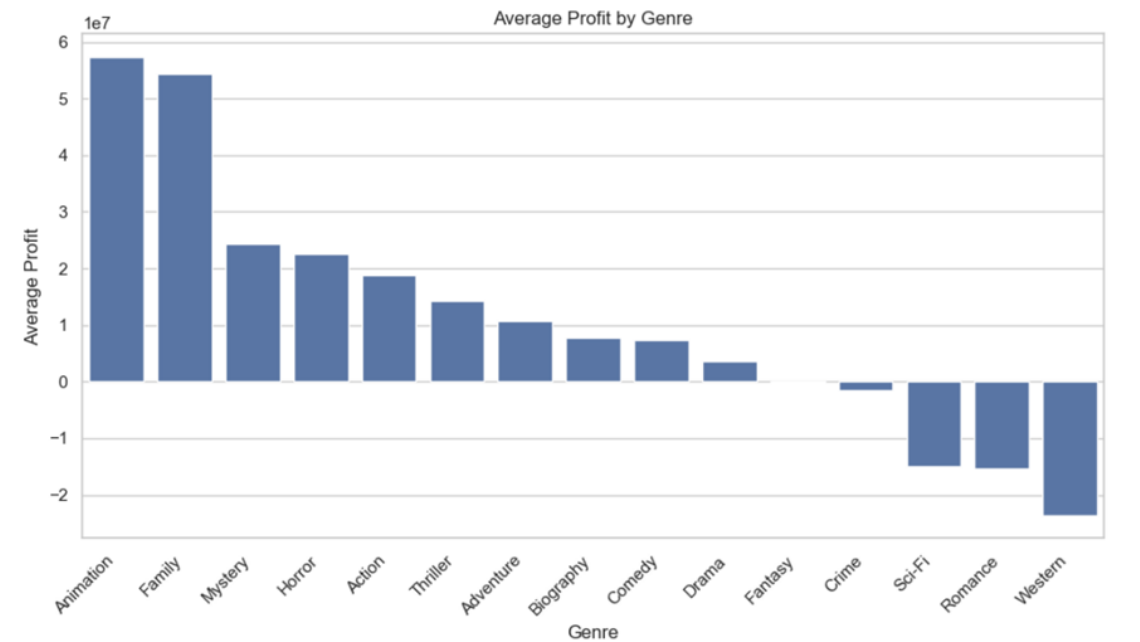
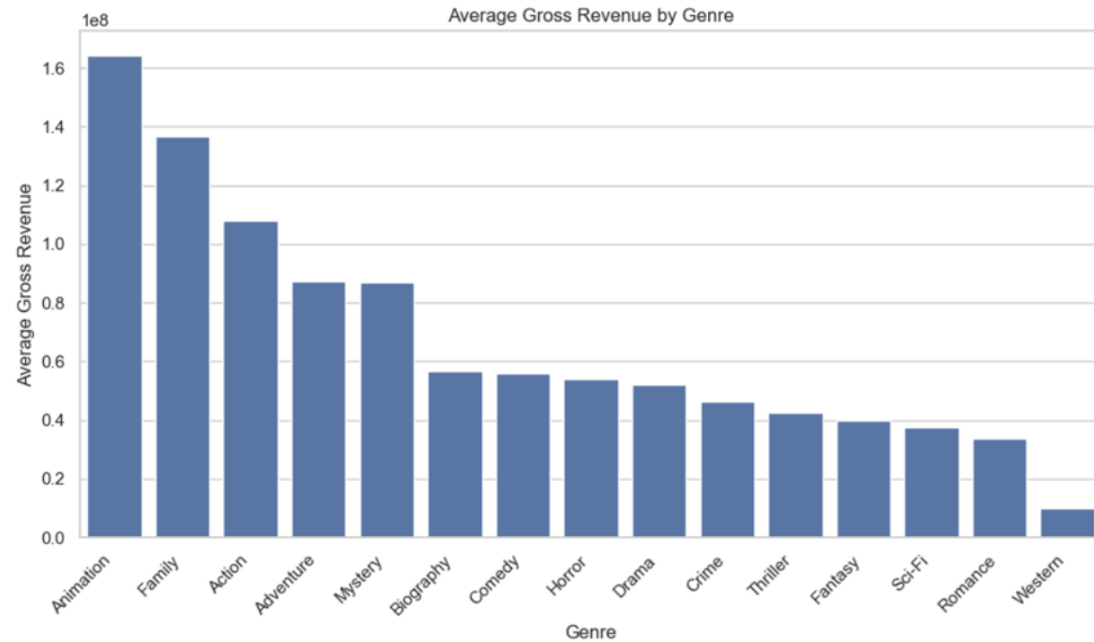
|                           | budget | budget_inflation_adjusted | gross | profit | \ |
|---------------------------|--------|---------------------------|-------|--------|---|
| budget                    | 1.00   | 0.89                      | 0.71  | 0.30   |   |
| budget_inflation_adjusted | 0.89   | 1.00                      | 0.67  | 0.14   |   |
| gross                     | 0.71   | 0.67                      | 1.00  | 0.80   |   |
| profit                    | 0.30   | 0.14                      | 0.80  | 1.00   |   |
| score                     | 0.07   | 0.03                      | 0.22  | 0.28   |   |
| votes                     | 0.51   | 0.45                      | 0.73  | 0.63   |   |
| runtime                   | 0.33   | 0.34                      | 0.27  | 0.11   |   |
| movie_name_length         | 0.13   | 0.10                      | 0.12  | 0.10   |   |
| years_since_production    | -0.32  | -0.10                     | -0.32 | -0.36  |   |

|                           | score | votes | runtime | movie_name_length | \ |
|---------------------------|-------|-------|---------|-------------------|---|
| budget                    | 0.07  | 0.51  | 0.33    | 0.13              |   |
| budget_inflation_adjusted | 0.03  | 0.45  | 0.34    | 0.10              |   |
| gross                     | 0.22  | 0.73  | 0.27    | 0.12              |   |
| profit                    | 0.28  | 0.63  | 0.11    | 0.10              |   |
| score                     | 1.00  | 0.51  | 0.43    | -0.02             |   |
| votes                     | 0.51  | 1.00  | 0.35    | 0.04              |   |
| runtime                   | 0.43  | 0.35  | 1.00    | 0.01              |   |
| movie_name_length         | -0.02 | 0.04  | 0.01    | 1.00              |   |
| years_since_production    | -0.06 | -0.32 | -0.09   | 0.01              |   |

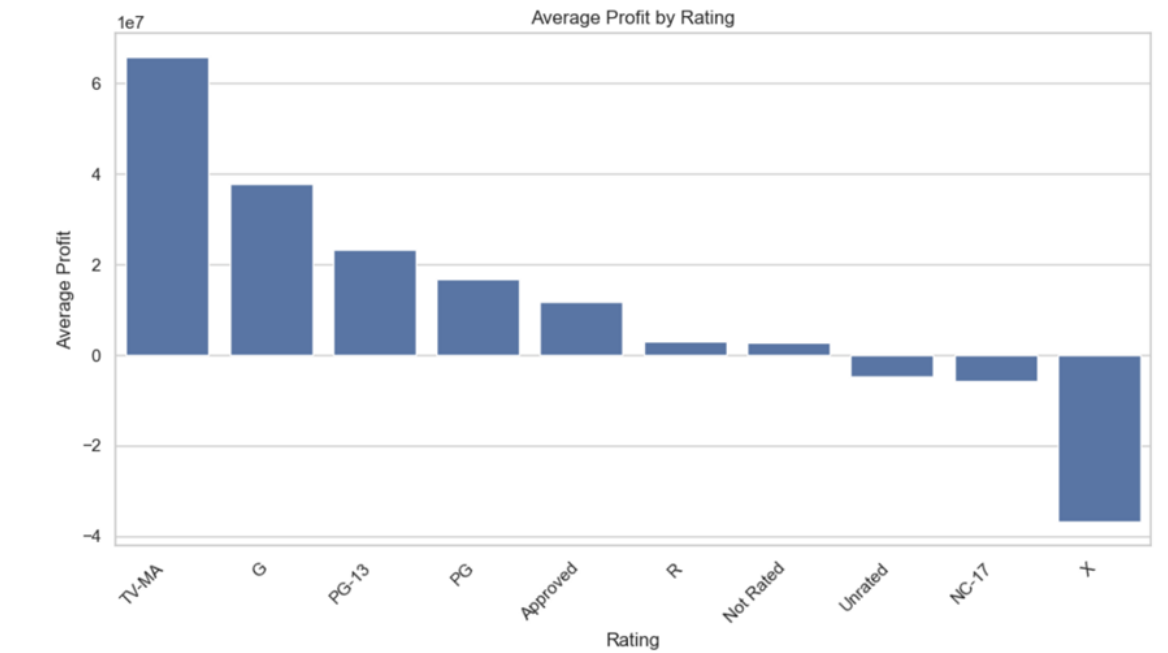
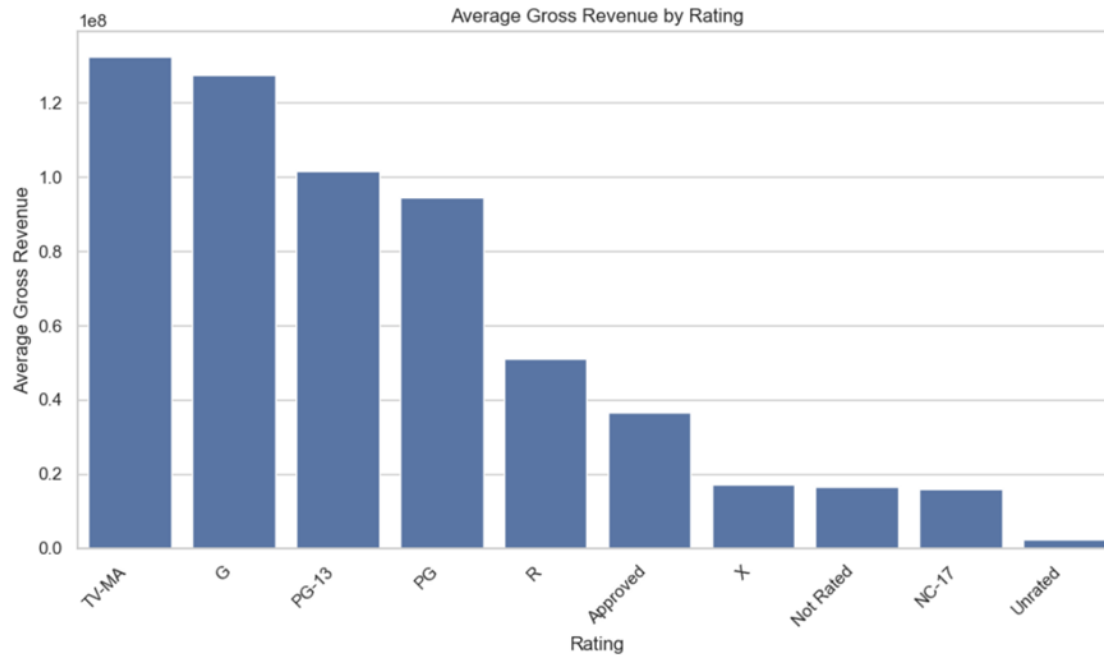
|                           | years_since_production |
|---------------------------|------------------------|
| budget                    | -0.32                  |
| budget_inflation_adjusted | -0.10                  |
| gross                     | -0.32                  |
| profit                    | -0.36                  |
| score                     | -0.06                  |
| votes                     | -0.32                  |
| runtime                   | -0.09                  |
| movie_name_length         | 0.01                   |

- **Key Descriptive Insights**
- What Drives Profit? (At a High Level)
- Profit is most strongly driven by audience engagement and gross revenue, not budget or ratings alone.
- **Strongest Correlation with Profit:**
- IMDb Votes ( $r = 0.63$ )  
→ Movies with more audience engagement earn much higher profits.
- **Very Strong Link:**
- Gross Revenue ↔ Profit ( $r = 0.80$ )  
→ Higher box-office revenue almost always leads to higher profit.
- **Other Key Relationships:**
- Score ( $r = 0.28$ ) → Better-rated movies earn moderately higher profits
- Budget ( $r = 0.30$ ) → Larger budgets increase revenue but do *not* guarantee profit
- Runtime ( $r = 0.11$ ) → Runtime has almost no meaningful impact on profit
- Years Since Production ( $r = -0.36$ ) → Older films earn significantly less profit
- **Takeaway:**
- Audience engagement matters more than spending.  
Campaigns that increase visibility, reviews, and votes have the biggest payoff.

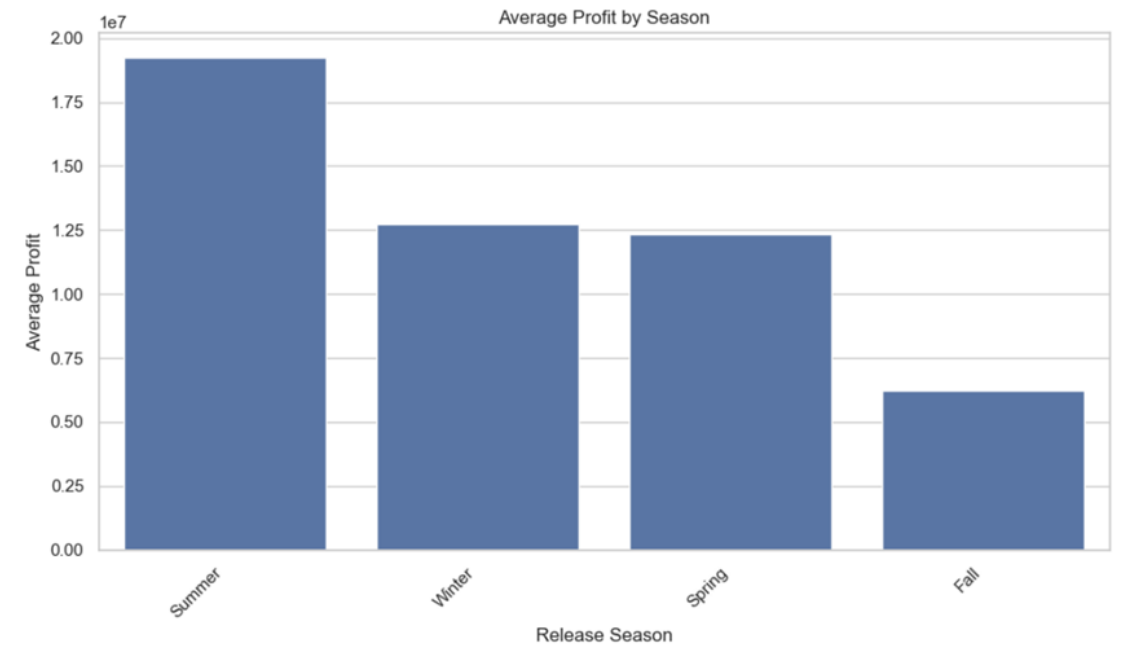
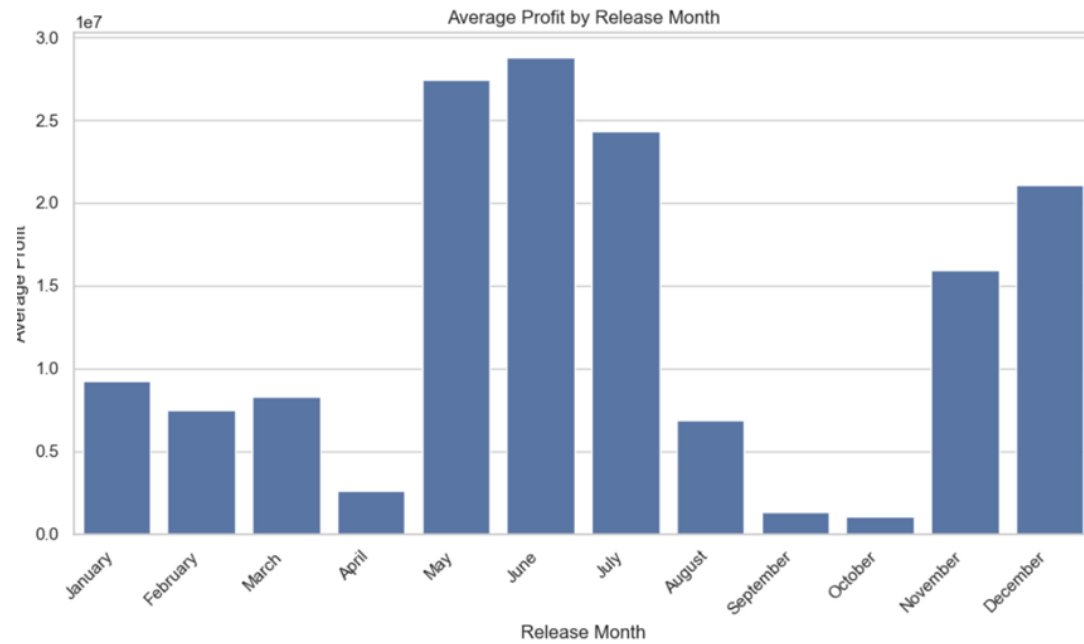
# Genre Insights



# Rating Insights

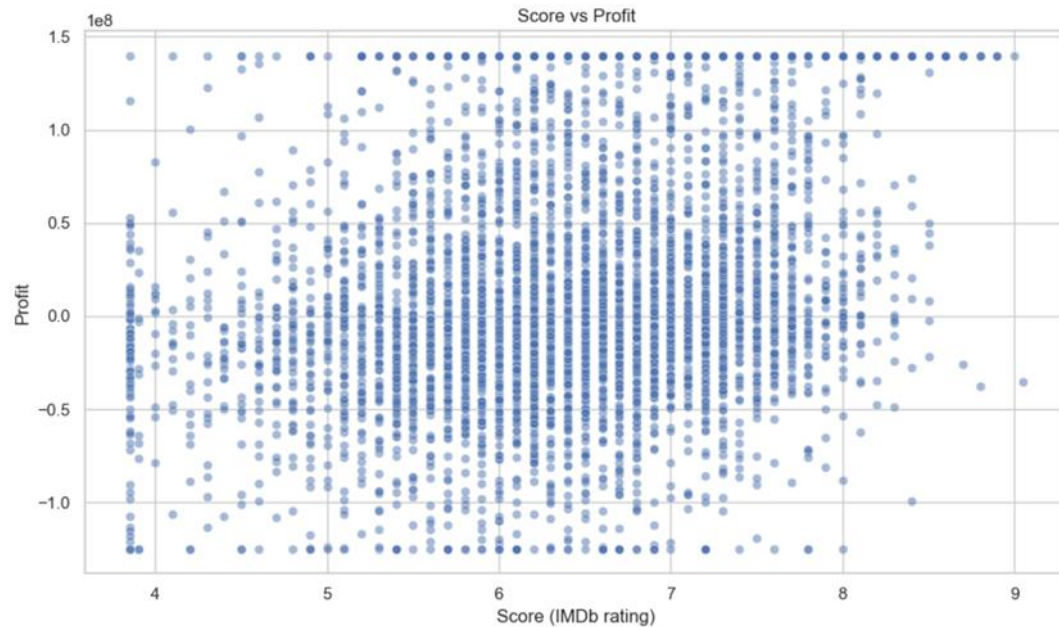


# Timing Insights

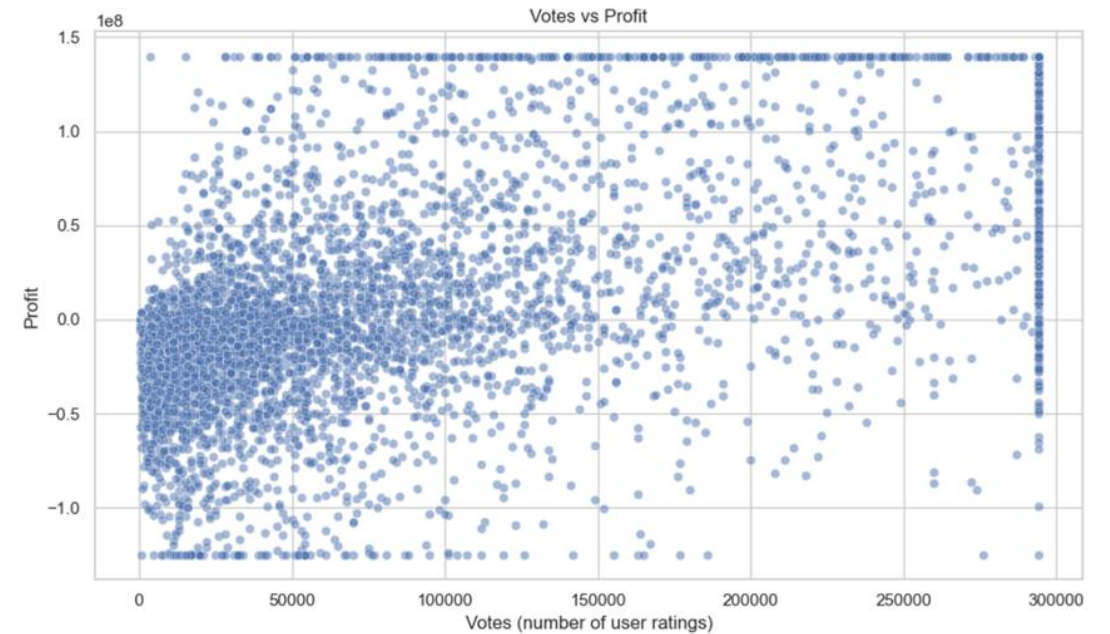




# Key Relationships



Correlation between score and profit:  $r = 0.28$



Correlation between votes and profit:  $r = 0.63$

# Strategic Recommendations

- **To Maximize Gross Revenue**
- **Prioritize high-earning genres:**  
*Action, Adventure, Animation*
- **Choose a PG-13 rating**  
Broadest audience reach and highest average gross revenue.
- **Release during peak windows:**  
*June–July* and *December* consistently show highest revenue.
- **Boost audience engagement early**  
Marketing that increases awareness/votes → higher gross.
- **Partner with strong distributors**  
Major studios amplify visibility and global revenue.

# Conclusion

- **Key Takeaways**
- Audience engagement (IMDb votes) is the strongest driver of profitability  
Movies with higher engagement consistently earn more profit ( $r = 0.63$ ).
- Genre and timing matter more than budget  
Animation, Family, and Comedy perform best; Summer and December releases yield the highest returns.
- Bigger budgets do not guarantee profitability  
Spending more increases gross revenue but does *not* ensure positive profit margins.
- Movie ratings influence reach but are not strong predictors of profit  
PG-13 maximizes audience size, but profit ultimately depends on genre and engagement.
- Predictive models offer direction, not certainty  
The Decision Tree performed best ( $R^2 = 0.52$ ), helping estimate profit but not perfectly forecasting outcomes.
- Data-driven decisions improve strategic planning, but creativity remains essential  
Combining analytics with strong storytelling gives producers the highest chance of success.